



Swedish Competence
Centre for Satellite-Enabled
Social Science Analytics



Rymdstyrelsen
Swedish National Space Agency

Beginner Earth Observation Analysis

Hand-on Workshop: From Flood Mapping to
Critical Infrastructure Accessibility



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SESAC, Project Assistant

25 September 2026 | Lund University

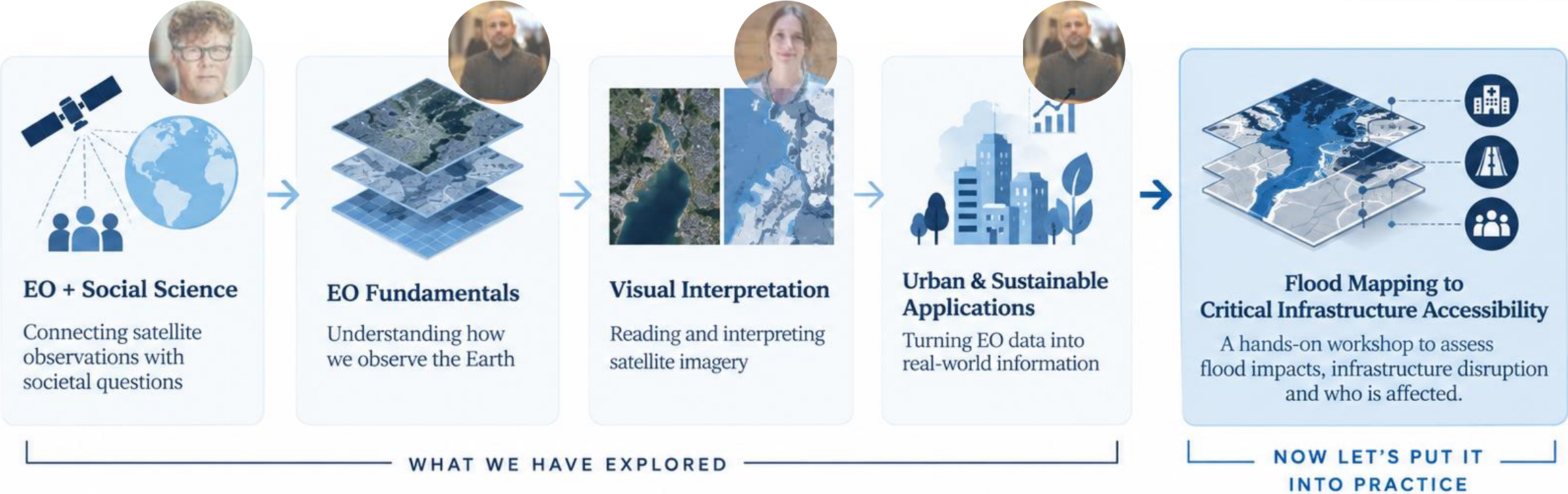


Scan to access
the materials



For EO fundamentals to real world applications

Building on what we have already learned



Let's start with you

Mentimeter activity #01



When you think about
flood impacts,
what comes to mind first?

Enter 1–3 words or short phrases.

There are no right or wrong answers.

 Mentimeter



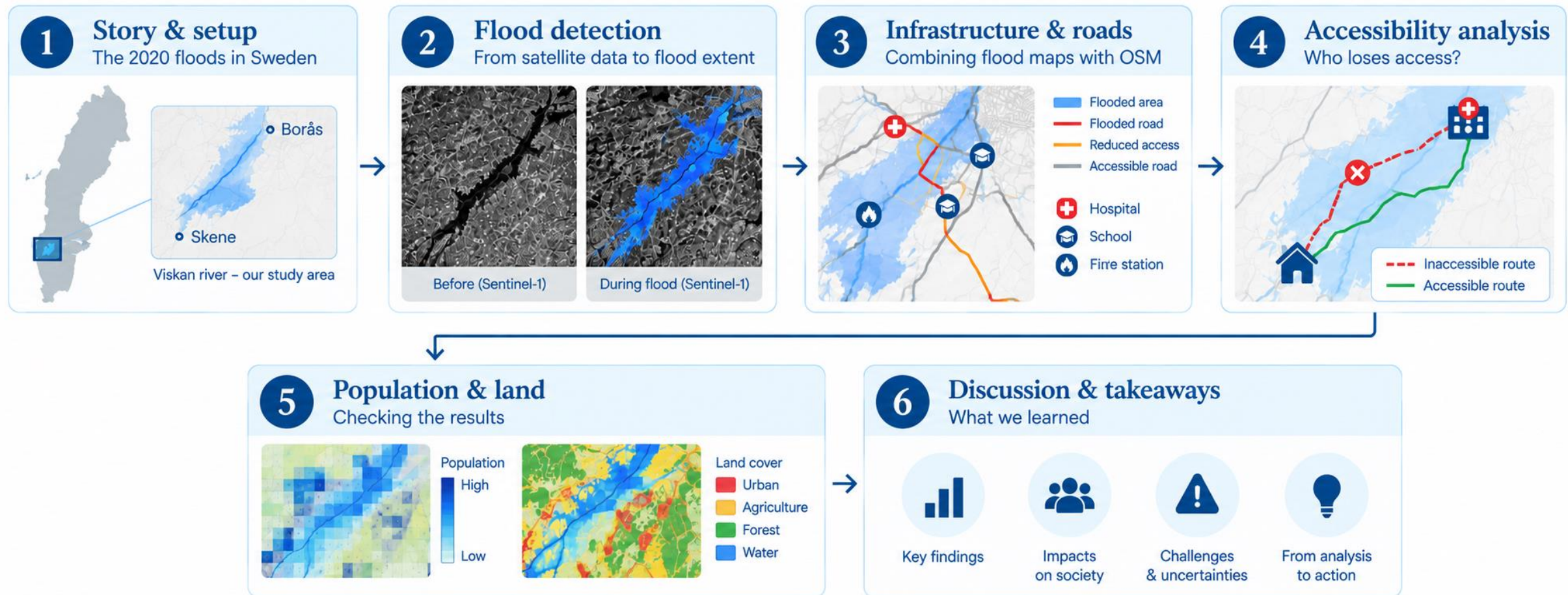
Go to **menti.com**

Enter the code

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Workshop overview

From satellite signal to societal applications



The 2020 floods

A large-scale flood event in southern Sweden



18 February 2020

Copernicus Emergency Management Service activated (EMSR427)



12 areas • ~ 41,200 km²
mapped across southern Sweden

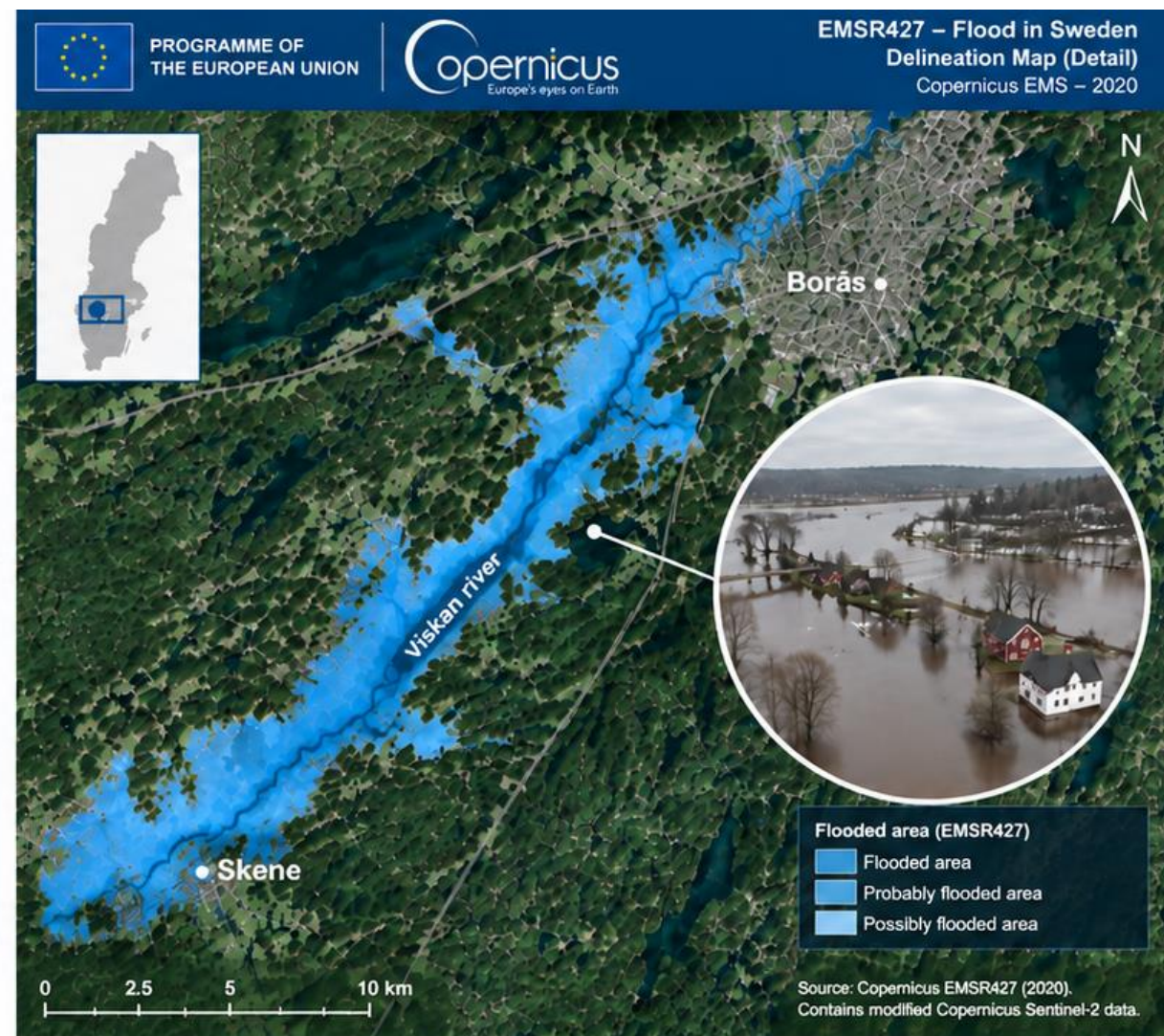


Sjuhärad / Viskan river

Borås → Skene
Our focus today

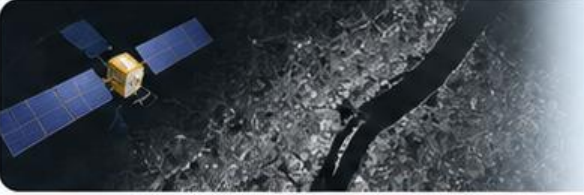


The Viskan river was **one of the two rivers** that EFAS flagged with the **highest flood risk** in this event.



Data sources

Real world data for real world impacts

DATA SOURCE		PROVIDER	ACCOUNT NEEDED?
	Sentinel-1 GRD SAR (VH)	 Copernicus Data Space Ecosystem (Sentinel Hub)	Yes
	Roads + hospitals/fire/police/schools	 OpenStreetMap (osmnx)	No
	Population grid	 SCB (Statistics Sweden) 1 km grid — GitHub repo	No
	Land cover raster	 NATUR VÅRDS VERKET NMD2018 (Naturvårdsverket) — GitHub repo	No



The population and land-cover rasters are already pre-clipped for our study area and hosted on GitHub, so no account is needed and no live download is required.

Creating a Sentinel Hub Account

One quick setup before we start working with



01

Create an account

Go to the Copernicus Data Space Ecosystem and sign up (or log in if you already have an account).

 PROGRAMME OF THE EUROPEAN UNION

 Europe's eyes on Earth



Copernicus Data Space Ecosystem

Explore. Process. Share.
A federated environment for Earth observation data services and applications.

Sign up

Log in

 <https://dataspace.copernicus.eu>

02

Open Sentinel Hub

After logging in, go to the Sentinel Hub Dashboard.

 Data Space Ecosystem

 Home

 Search

 Browse

 Data processing

 Analytics

 Applications

 Sentinel Hub

 Documentation >

03

Create an OAuth client

In your Sentinel Hub account, go to User Settings → **OAuth clients** and click Create.



Dashboard

Configuration

Usage

Billing

User Settings

User Settings

Profile

OAuth clients

Sessions

+ Create OAuth client

Name *

SESAC_Workshop

Create

04

Save your credentials

After creating the client, copy your Client ID and Client Secret. You will use these in the notebook.

OAuth client created

 Your OAuth client was created successfully. Make sure to copy your Client Secret now. It will not be shown again.

Client ID

.....



Client Secret

.....



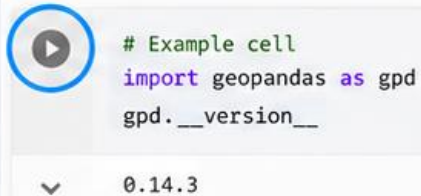
Done

Using Google Colab Notebook

A few quick tips to get the most out of the hands-on session

1 Run cells top to bottom

Click the ▶ button on the left of each cell, or press Shift + Enter.



Run the cells in order and don't skip ahead.

2 You won't write any code

All the code is already prepared. Your job is to run the cells and read the output.



Focus on the results, figures and maps.

3 You may need to restart the runtime

The first cell installs the required packages. After this, Colab will ask you to restart the runtime — this is normal!

Restart runtime?

To apply the installed packages, Colab needs to restart the runtime. Your variables will be lost.

Cancel

Restart runtime



Click restart, then continue from the next cell.

4 Look for Activity boxes

Grey boxes labeled “Activity” are optional discussion prompts, not extra tasks.



Activity

*Can you see differences between the two images?
What might be causing them?*



Read, think for a moment, and discuss if you like — then keep going.

5 Need help?

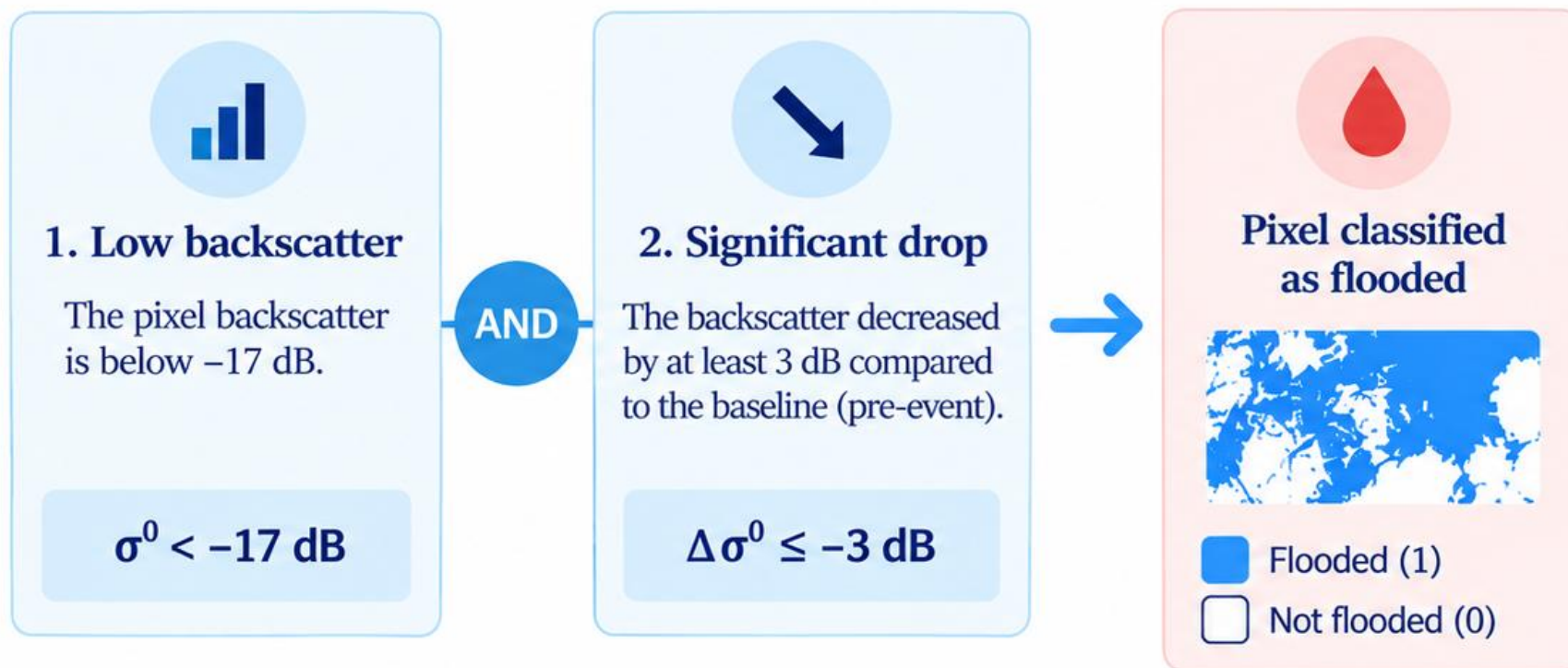
If you get stuck on a cell, raise your hand and ask for help. Don't try to guess or scroll ahead.



Stefanos and Sai will be around the room to help with any questions.

How we detect flooding from Sentinel-1

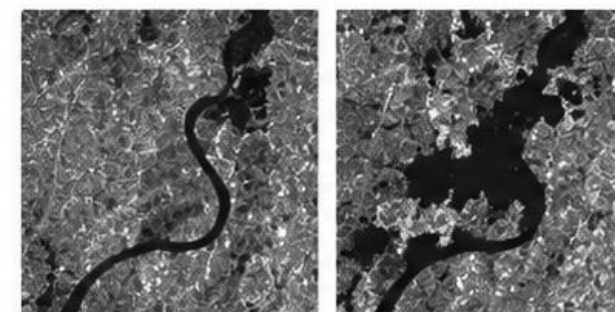
A simple and robust rule



Why both conditions?

Using both an absolute threshold and a significant drop ensures that we detect new flooding, rather than permanent water bodies (e.g. lakes or large rivers) that always appear as dark in SAR images.

Example (Sentinel-1 backscatter)



Before (baseline)

During flood

Detected flood (binary mask)



What this method can (and cannot) tell us

Strengths and limitations of our flood detection formula



What it can tell us



Detects where water is present

Sentinel-1 SAR is sensitive to open water surfaces, allowing us to map inundated areas.



Identifies newly flooded areas

By comparing pre- and post-event images, we detect areas that became water-covered.



Provides consistent, large-scale coverage

Works under cloudy conditions and over large areas.



Useful for understanding impacts on the road network

Helps us identify which road segments are affected and may become unusable.



What it cannot tell us



It does not provide water depth

SAR measures surface reflectance, not the height or depth of the water.



We do not use terrain information (e.g. slope/DEM filtering)

Some detections may occur on flat, low-lying areas that are normally wet (e.g. permanent water bodies).



A road is treated as either completely unavailable or fully available

We do not model partial obstruction (e.g. shallow or muddy water). The classification is binary: flooded or not flooded.



Some static water bodies may still appear as flooded

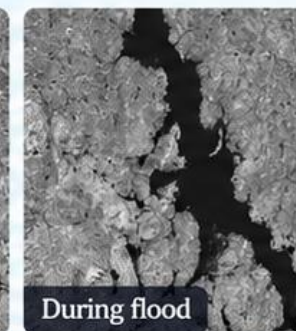
Although we use a change detection approach, very small or temporal changes in backscatter can occasionally lead to false detections.

In this workshop, we use a simplified, binary representation of flooding, based on Sentinel-1 observations.

Sentinel-1 backscatter (VV)

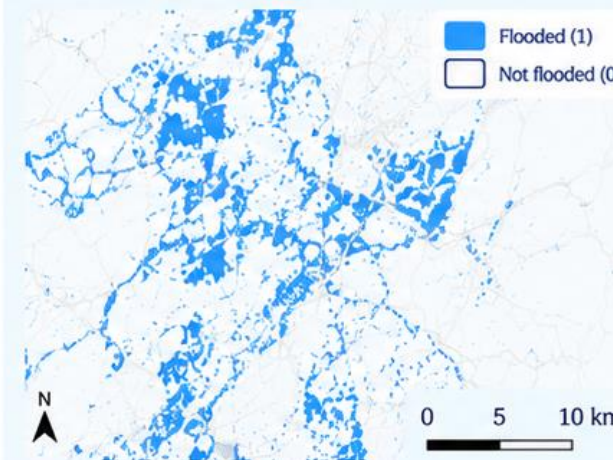


Before (baseline)



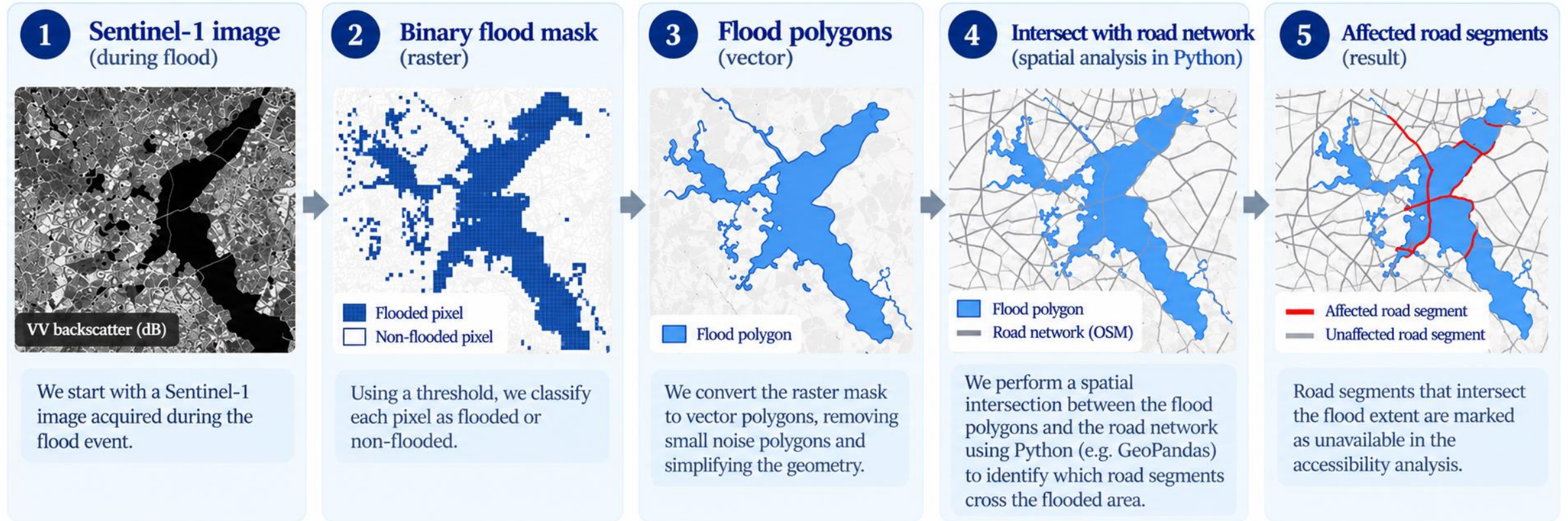
During flood

Detected flood extent



From pixels to network disruption

Turning an EO observation into something we can analyse in Python



Key takeaway

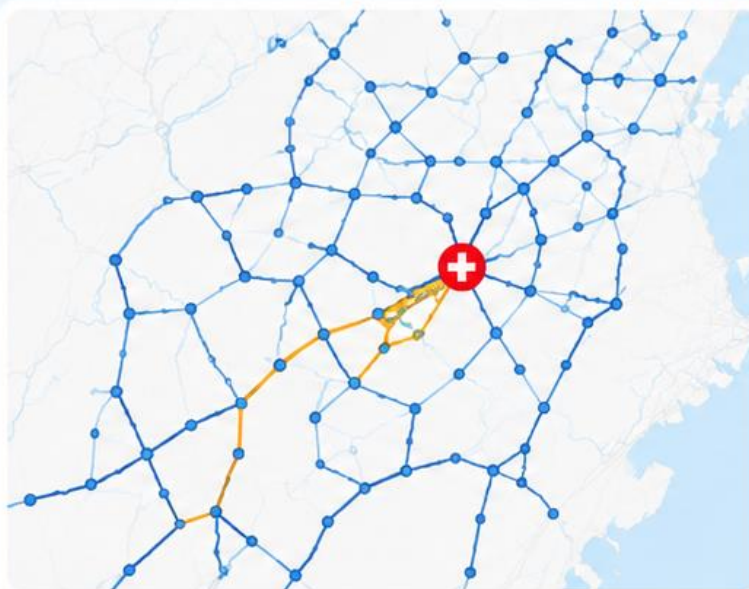
The satellite result is initially a raster of flooded pixels, while roads are vector geometries. By converting the flood mask to polygons, we can combine the two in Python and identify which road segments are affected.

From road to accessibility

Graph construction and isochrones

1 Graph construction

We turn the road network into a graph.

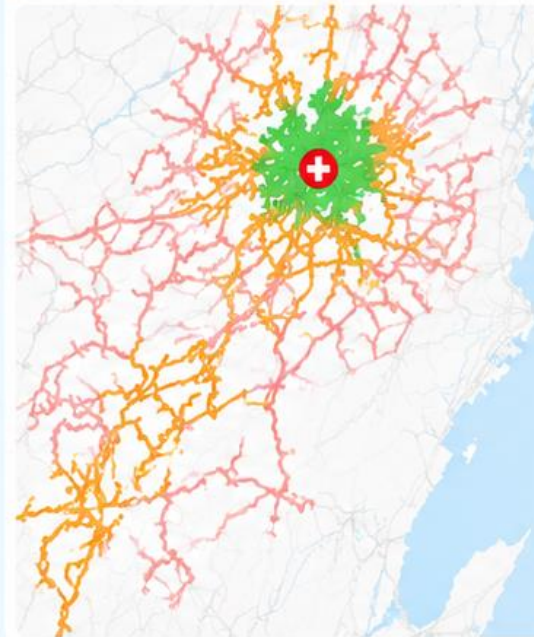


- Road node
- Road edge (connection)
- Example route
- ⊕ Hospital

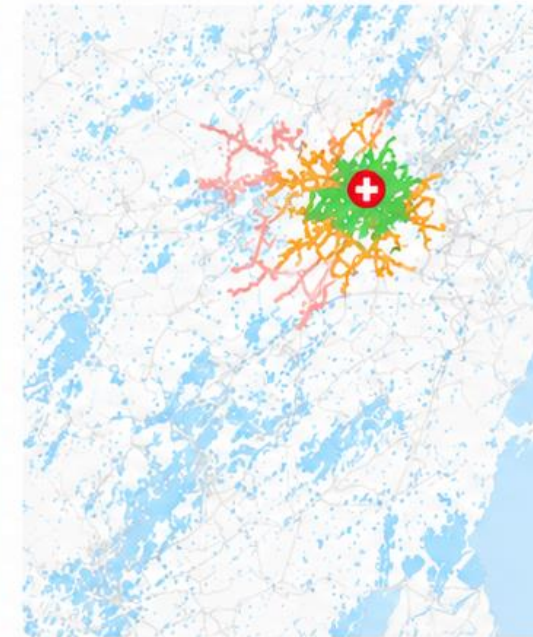
2 Isochrones (travel time areas)

We compute the area reachable within a given travel time from each hospital.

Normal conditions



During flood (Sentinel-1)



Travel time isochrones

- 5-min drive
- 10-min drive
- 15-min drive
- Flooded area (Sentinel-1)
- ⊕ Hospital

Let's hear from you

Mentimeter activity #02



Rank these by expected
accessibility impact —
biggest drop first.



Hospital



Fire station



School



Mentimeter



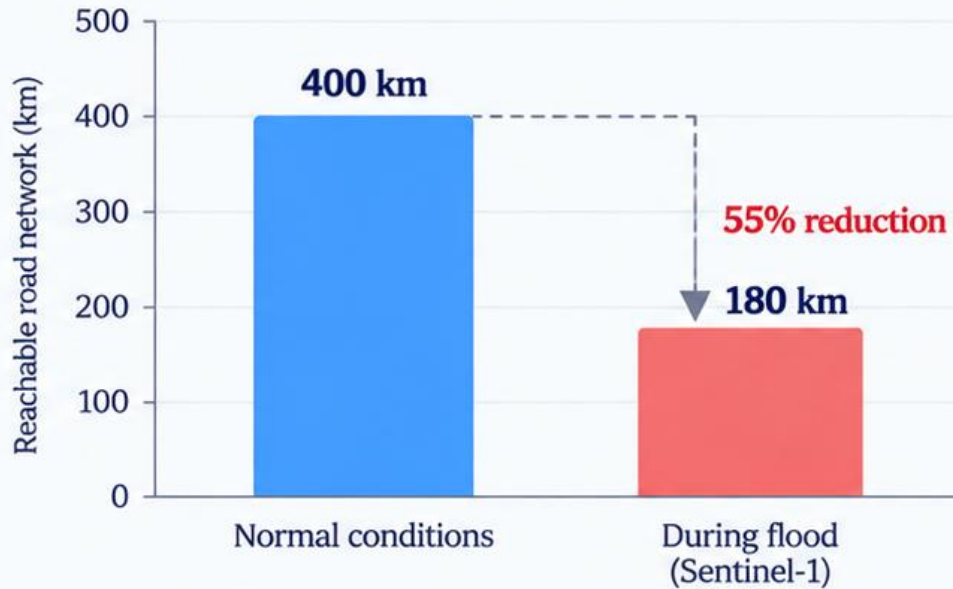
menti.com

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Quantifying the impact on accessibility

Accessibility reduction formula

Example: 10-min accessibility to Hospital



Formula

$$\text{Accessibility reduction (\%)} = \frac{L_{\text{normal}} - L_{\text{flood}}}{L_{\text{normal}}} \times 100$$

L_{normal} = total length of the reachable road network within the selected travel time under normal conditions (km)

L_{flood} = total length of the reachable road network within the selected travel time during flood conditions (km)

How to interpret the results



0%

No reduction

The reachable network is the same as in normal conditions.



~100%

Almost no access

Very little (or none) of the network remains reachable.



Key takeaway

The accessibility reduction tells us how much the reachable road network is affected by flooding. A higher percentage means a stronger impact on access to the facility, even if the facility itself is not directly flooded.

From accessibility to exposure

Adding the human and environmental layers

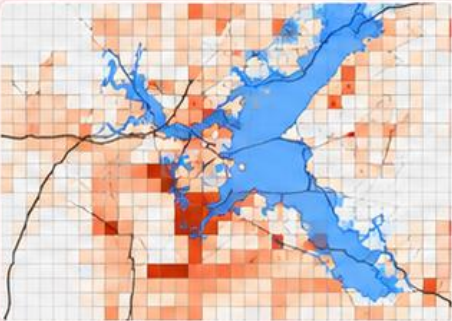
Flood extent and disrupted access are important, but the key question is:
Who and what is affected?



Population

Who is affected?

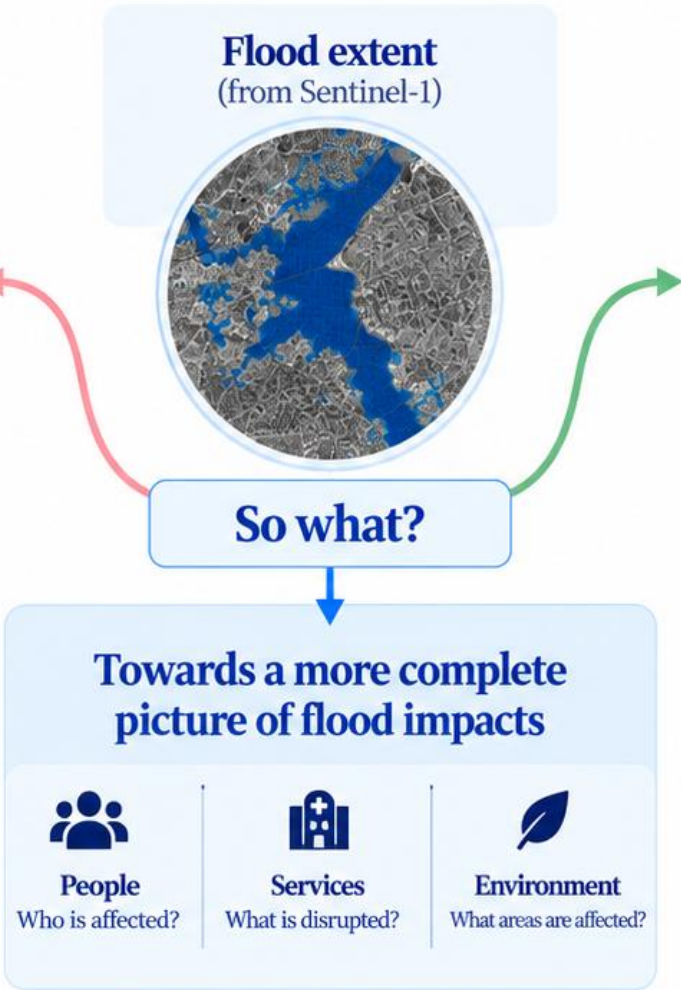
- **Direct exposure**
How many people live within the flood extent?
- **Accessibility exposure**
How many people lose previously available access to a hospital (e.g. within 10 minutes)?
- **Different travel times**
Compare 5, 10 and 15 minutes.



Population (people per cell)
1 – 10
11 – 50
51 – 100
101 – 500
> 500

■ Flood extent

This helps us move from kilometres of affected roads to people who may be impacted.

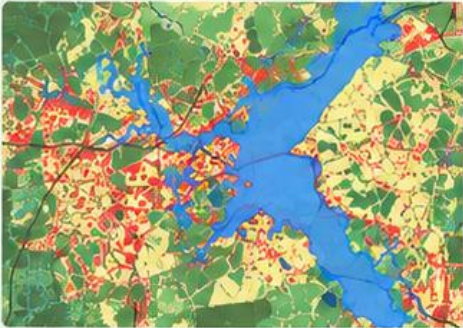




Land cover

What environment is affected?

- **Which land cover classes are flooded?**
e.g. urban areas, agricultural land, forests, wetlands, water bodies.
- **Which classes are disproportionately affected?**
Understand the environmental context of the flooding.
- **Where might SAR perform less well?**
Some surfaces (e.g. vegetation, urban areas) can lead to underestimation of flooding.



Land cover class (example)
Urban
Agriculture
Forest
Wetlands
Water

■ Flood extent

Land cover provides environmental context and also helps assess the limitations of the Sentinel-1 detection.



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Thank you! 😊

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GIS, and spatial analysis through lectures,
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WORKSHOP-ONLY REG. DEADLINE: 11 SEP



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